Nitesh Gokul

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.95
FIITJEE JUNIOR COLLEGE, NARAYANAGUDA	Class XII (Telangana State Board)	2023	96.20%
FIITJEE WORLD SCHOOL, NARAYANAGUDA	Class X (SSC)	2021	10.00%

EXPERIENCE

Applied Research Intern - Thomson Reuters Labs

Research intern building and evaluating an in-product LLM agent for statutory report editing at Thomson Reuters.

- Engineered **In-Product Agent** automating **Statutory Report** editing by combining natural language understanding with report-generation pipelines, enabling plain-language report edits without manual template navigation.
- Optimized agent accuracy and latency through prompt refinement, precompute caching, and evaluation bug fixes, lifting pass rate by up to **8.1%** and cutting latency by up to **38%** across three evaluation suites.
- Integrated user-proxy agent into evaluation framework to simulate realistic multi-turn user interactions, including option selection and clarification responses, enabling fully automated, scalable behavioural testing without manual intervention.
- Investigated model-level regressions, including reasoning-mode incompatibilities and silent service crashes under concurrent load, to keep evaluation results reliable as underlying language model versions changed.

PROJECTS

GeoNLI: Natural Language Interpretation of Satellite Imagery - Inter IIT Tech Meet 14.0:

- Architected a Service-Oriented pipeline decoupling satellite imagery interpretation as part of Inter IIT Tech Meet 14.0 using **YOLOv11-OBB**(oriented detection), **SAM 3**(mensuration), and a fine-tuned **Qwen2-VL-7B**(semantic reasoning).
- Designed an **Attribute Lattice** with lazy evaluation for efficient state management, and a dynamic **LoRA adapter-switching** mechanism to preserve instruction, following consistency across multi-step reasoning during inference.
- Containerized and integrated individual components into a single, reproducible end-to-end pipeline using Docker

AdherenceGuardian:

- Architected a multi-agent **LangGraph** system with coordinated agents (planning, monitoring, barrier resolution, liaison) that models medication scheduling as an optimization problem balancing drug-interaction safety and dosing timing.
- Integrated a **ChromaDB** based **RAG** pipeline over clinical knowledge bases (RxNorm, DrugBank, SIDER) with time-series anomaly detection for adherence monitoring, improving simulated adherence from 65% to 90% over 30 days.

Ad Click Prediction Model:

- Designed a click-through rate prediction model for the **American Express Decision Track** competition, using user data to estimate ad-click likelihood and enhance ad targeting strategies.
- Achieved a Mean **Average Precision(MAP)** score of 0.440 with a **LightGBM** model and 0.439 with a neural network trained on top 100 **SHapley Additive exPlanations(SHAP)**-selected features.

Anchor-Guided Dual-Memory Propagation for OCT Segmentation:

- Collaborated with ophthalmology faculty at the University of Pittsburgh to ground the framework in clinical annotation practices and validate segmentation results against expert-graded OCT scans. Paper submitted for IEEE HealthCom'26.
- Developed a a bidirectional **memory-propagation** framework on a PVT-v2 backbone to trace Subretinal

Hyperreflective Material (**SHRM**), an **AMD** biomarker, across 3D **OCT** volumes from clinician-labeled slices.
- Combined a locked reference-memory bank of clinician-verified slices with a frozen U-Net, lifting SHRM segmentation accuracy from **0.466** to **0.677** IoU and **0.584** to **0.794** Dice while labeling under **9%** of slices per volume.
- Showed via ablation that removing the reference-memory bank collapsed accuracy to **0.455-0.504** IoU, confirming it as the key mechanism preventing error drift across long unannotated gaps between clinician-labeled slices.

POSITIONS OF RESPONSIBILITY

Tech Lead - Reinforcement Learning at Epoch(AI/ML Club)

Core at Epoch (AI/ML Club, IITH)

Assistant Manager in Media & Publicity at E-Cell IITH:

Campus Coordinator for **Prosperity'26** conducted by **IMC Trading**

Campus Coordinator for **Integral Cup'26** round 1 conducted by **STEMvibe**

SCHOLASTIC ACHIEVEMENTS

Secured **AIR 30** in **Prelims Juniors Squad** among **10k+** teams and invited to IIT Guwahati for **Technothlon 2019**

Secured **Runners Up** in **Innov-AI-thon,** Elan&nVision'26 for building an AI based solution in medical field.

Secured **2nd-Runner Up** in **Dev Duel,** Elan&nVision'26 for building an agentic platform for medical adherence.

Secured **2nd-Runner Up** in **EnigmaCTF** a Capture The Flag event conducted by Elan&nVision'26.

SKILLS

Languages: C/C++, Python, HTML, CSS, JavaScript.

Frameworks: PyTorch, Tensorflow, Transformers, Langchain, Langgraph, LlamaIndex, OpenCV, Seaborn, ChromaDB, OpenAIGymnasium, NTLK, Sci-Kit Learn, Matplotlib, Pandas, NumPy, ReactJS, NextJS, NodeJS.

Tools: Git/GitHub, LaTeX, MySQL, React, Vite.

Technical Skills: Machine Learning, Reinforcement Learning, Computer Vision, Natural Language Processing, Deep Learning, Algorithmic Optimisation, Product Develpment, Web Development.

RELEVANT COURSES

CV, NLP, RL, DL, Online Learning, Programming for AI, Foundations of Machine Learning, Robotics, Probability and Random Variables, DSA, Algorithms, Convex Optimization, Fraud Analytics, Predictive Modelling, Discrete Mathematics, Matrix Theory, Computer Architecture, Statistics, DBMS, OS, Compilers.